

Module 7 – Manual Agent with Tool Calling

LLM Integration

Connect a local LLM through Ollama, send prompts, and receive responses. Keep model configuration separate from logic. This is the foundation for the agent.

Key parameters: model, messages, temperature

System Prompt

Defines assistant behavior and tool usage rules. Reduces hallucinations and keeps answers consistent. Guides the model during reasoning.

Key parameters: role, rules, constraints

Tool Registry

Maintains all available tools and their metadata. Exposes tool definitions to the LLM. Makes adding tools easy.

Key parameters: name, description, parameters

Tool Executor

Finds the requested tool and executes it safely. Returns structured output and handles failures. Keeps execution isolated.

Key parameters: tool_name, arguments, result

Agent Loop

Calls the LLM, checks for tool calls, executes tools, and sends results back. Continues until a final answer is produced. Central orchestration component.

Key parameters: messages, tool_calls

Weather Tool

Uses Open■Meteo to retrieve live weather. Converts city to coordinates before requesting data. Returns structured JSON.

Key parameters: city, latitude, longitude

Browser Tool

Downloads web pages and extracts readable text. Removes scripts and styles before parsing. Supplies page content to the LLM.

Key parameters: url, timeout

Web Search Tool

Searches the web and returns top matching results. Provides title, URL, and snippets. Used for up-to-date information.

Key parameters: query, max_results

SQL Tool

Executes database queries against a configured database. Returns rows in a structured format. Useful for data-driven answers.

Key parameters: connection, sql, result

Architecture

